When Can Internal-State Adaptation Replace Weight Adaptation in Vision–Language Models?

Anonymous Authors

***Abstract*—Few-shot adaptation of a vision–language model (VLM) commonly changes model weights, even when the target shift may require only a small correction to how visual evidence is read. We ask a broader question: *when can adapting a frozen model's internal state replace weight adaptation?* We study a simple test-time method, *Key–Value Test-Time Tuning* (KV-TTT), that constructs a low-rank basis from the second moment of support correctness gradients and learns a bounded residual on task-relevant visual key/value activations. By the Ky Fan maximum principle, this basis is the rank-r subspace retaining the maximum support gradient energy. Across disaster assessment, histopathology, robot failure recognition, and vision–language action classification, we find that energy coverage alone is not predictive. Successful internal-state adaptation additionally requires signed support–query directional transfer and an activation site with sufficient functional authority. On fixed- semantics BRIGHT, KV-TTT exceeds matched LoRA on average with $1{,}024$ learned coefficients versus 10.1M LoRA parameters, and the pattern replicates across VLM families. Camelyon exhibits stable geometry and a query-basis KV oracle that reaches LoRA, while RoboFail exposes class-mixture and site mismatch: oracle prior reweighting raises directional agreement from $0.209{\pm}0.361$ to $0.934{\pm}0.025$. ManipBench remains far below LoRA even with query-derived bases and broader activation sites, identifying a capability boundary rather than an optimization failure. These results turn internal-state adaptation from a universal replacement claim into a testable account of when local, calibration-preserving adaptation is sufficient.**

***Index Terms*—test-time adaptation, vision–language models, internal-state adaptation, gradient geometry, parameter-efficient adaptation.**

I. Introduction

Few-shot deployment poses an ambiguous adaptation problem. A new domain may preserve a capability already present in a VLM and change only which visual evidence should be trusted. Alternatively, it may require a new language–vision decision rule or action mapping. Weight-space methods such as LoRA [1] can address both cases, but use far more degrees of freedom than necessary in the first and can produce severely overconfident errors. Internal-state adaptation is a compelling smaller alternative, yet its applicability conditions are poorly understood.

We ask whether event adaptation can instead be expressed as a low-dimensional change in a frozen model's working activations. Transformer key/value projections form a natural intervention point [2], [3]. For each labeled support example, we differentiate the answer-token loss with respect to the post-event image-token projection outputs. Crucially, we retain the second moment $\sum_i G_i^\top G_i$, not the average gradient. Its leading eigenvectors preserve a space of corrective directions even when example-level gradients cancel in the mean.

Our central hypothesis is that successful internal-state test-time adaptation requires two conditions: *directional geometry transfer*, meaning that support and query request compatible corrections in the learned subspace, and *actuator sufficiency*, meaning that the intervened activation can realize the target functional change. We test this hypothesis rather than expanding the method until it approximates a small weight adapter.

Our contributions are: (i) a frozen-backbone, visual-KV residual whose gradient-covariance basis is optimal for rank-constrained support energy retention; (ii) energy-overlap (ρ), signed-alignment (κ), class-conditional, query-basis, and projection-site diagnostics that separate geometry from actuation; (iii) evidence across BRIGHT, Camelyon17, Guardian/FailCoT, and ManipBench, including multiple VLM families and matched three-seed task evaluations; and (iv) explicit positive, mismatch, and capability-boundary regimes. Query labels are used only in clearly marked mechanism oracles, never for primary adaptation or model selection.

II. Related Work

Test-time adaptation. Test-time training [4], TENT [5], CoTTA [6], and test-time prompt tuning [7] adapt normalization, objectives, or prompts at deployment. KV-TTT instead uses a small labeled event support once, produces a fixed event controller, and adds no query context.

Parameter-efficient and activation-space adaptation. LoRA and adapters modify weights [1], [8]; prompt tuning learns input-like parameters [9]. Transformer layers have also been interpreted and edited as key–value memories [3], [10], [11]. Our method neither edits nor adds model weights: it applies a bounded residual to selected activation rows.

Remote-sensing damage assessment. xView2 [12] and BRIGHT [13] support building-level damage research, while remote-sensing foundation representations and VLMs broaden transfer [14], [15]. We focus on same-event, few-example adaptation of a general VLM under strict spatial separation.

III. Method

A. Gradient-covariance subspace

Let f_θ be a frozen Qwen2.5-VL-7B model [16]. For support example (x_i, y_i), projection type $t \in \{K, V\}$, and selected decoder layer l, let $Z_i^{(l,t)} \in \mathbb{R}^{T\times d}$ denote the projection output. A binary mask M_i selects only the second (post-event) image-token group. We compute the answer-span loss and masked gradient

$$G_i^{(l,t)} = \nabla_{[Z_i^{(l,t)}]_{M_i}} \mathcal{L}(x_i, y_i). \tag{1}$$

The uncentered gradient second moment and rank-r basis are

$$C^{(l,t)} = \frac{1}{N}\sum_{i=1}^{N}(G_i^{(l,t)})^\top G_i^{(l,t)}, \quad B^{(l,t)} = \mathrm{TopEig}_r(C^{(l,t)}). \tag{2}$$

This is equivalent to retaining the leading right singular directions of the stacked token gradients. It preserves directional variation that $\bar{g} = N^{-1}\sum_i g_i$ discards.

a) Optimal rank-constrained energy retention.: Let $C \succeq 0$ and let $B \in \mathbb{R}^{d\times r}$ satisfy $B^\top B = I_r$. The correctness-gradient energy retained by the subspace is

$$\mathcal{E}(B) = \mathrm{tr}(B^\top CB) = \frac{1}{N}\sum_i \|G_i B\|_F^2. \tag{3}$$

By the Ky Fan maximum principle [17],

$$B^* = \mathrm{TopEig}_r(C) \in \arg\max_{B^\top B = I_r} \mathrm{tr}(B^\top CB), \tag{4}$$

and the optimum equals the sum of the largest r eigenvalues of C. Thus the basis is not an arbitrary SVD heuristic: among all rank-r subspaces it preserves the most support correctness-gradient energy. This property is deliberately limited. It guarantees neither that query gradients occupy the same subspace, that their signed mean agrees with support, nor that a residual at the chosen activation can express the required decision change.

b) Geometry and actuator diagnostics.: For support basis B_S and stacked query gradients G_Q, we measure energy transfer and signed agreement as

$$\rho = \frac{\|G_Q B_S\|_F^2}{\|G_Q\|_F^2}, \qquad \kappa = \cos(P_{B_S}\bar{g}_S, P_{B_S}\bar{g}_Q), \tag{5}$$

where $P_{B_S} = B_S B_S^\top$. High ρ says that relevant query energy is covered; high κ says that the requested average correction is aligned. We then hold subspace size fixed and vary the intervention site or use a query-derived basis as an oracle test of actuator authority. These diagnostics read query labels and are never deployable performance estimates.

B. Bounded residual controller

For a Diagonal controller,

$$Z' = Z + M \odot \left[(ZB)\,\mathrm{diag}(\alpha \tanh a)B^\top\right]. \tag{6}$$

For a Full controller, a raw $r \times r$ matrix R gives

$$A = \frac{\alpha R}{1 + \|R\|_F}, \qquad Z' = Z + M \odot \left[(ZB)AB^\top\right]. \tag{7}$$

Thus $\|A\|_2 < \alpha$. All coefficients start at zero, so the controller initially implements the identity. We use the middle and final decoder layers (14 and 27); both K and V are controlled. Rank-16 Full and Diagonal variants learn $2\times2\times16^2 = 1{,}024$ and $2\times2\times16 = 64$ scalars, respectively. The compact rank-5 variants learn 100 and 20.

Algorithm 1 Supervised event-level KV-TTT

Require: Frozen VLM f_θ; support $\mathcal{S}$; layers $\mathcal{L}$; rank r; bound α; updates T_u

for $(x_i, y_i) \in \mathcal{S}$ **do**
  Build the paired-image prompt and isolate the answer-token span.
  Backpropagate its loss; retain post-image gradients $G_i^{(l,t)}$.
  $C^{(l,t)} \leftarrow C^{(l,t)} + (G_i^{(l,t)})^\top G_i^{(l,t)}$.
end for
for $l \in \mathcal{L}$ and $t \in \{K, V\}$ **do**
  $B^{(l,t)} \leftarrow \mathrm{TopEig}_r(C^{(l,t)})$.
  Initialize the Diagonal vector a or Full matrix R to zero.
end for
for $u = 1, \ldots, T_u$ **do**
  Accumulate the mean labeled loss over all $|\mathcal{S}| = 24$ items.
  Add $\lambda\|\phi\|_2^2$, clip gradients, and update ϕ.
end for
return Bases B and controller coefficients ϕ.

C. Coefficient fitting

The frozen VLM and bases remain fixed while Adam optimizes only controller coefficients on all 24 support examples:

$$\min_\phi \frac{1}{N}\sum_i \mathcal{L}(f_{\theta,B,\phi}(x_i), y_i) + \lambda\|\phi\|_2^2. \tag{8}$$

The common configuration uses four full-support updates, learning rate 0.05, $\lambda = 10^{-3}$, and unit gradient clipping. Each query then runs normally with the fixed controller; no query-time optimization is performed.

a) Complexity.: For L selected layers and both projection types, Diagonal learns $2Lr$ scalars and Full learns $2Lr^2$. Basis extraction requires one frozen-model backward pass per support item and an eigendecomposition of each $d \times d$ second-moment matrix. The controller adds two low-rank projections per selected module at inference but no extra tokens, retrieval, or optimization. Bases and coefficients are serialized together; their storage is dominated by B, not by the learned coefficient count.

IV. Experiments

A. Protocol

We use the four BRIGHT events for which the strict balanced protocol is complete: Hawaii wildfire, Libya flood, Noto earthquake, and Turkey earthquake. Each event has 24 same-event labeled support instances (8 per class) and 300 query instances (100 per class); support and query tile sets are disjoint. All arms start independently from raw `Qwen/Qwen2.5-VL-7B-Instruct`, use 448×448 paired crops, and score the same query IDs. The clean LoRA baseline uses rank 16, alpha 32, $q/k/v/o$ targets, learning rate 10^{-4}, batch size 1, gradient accumulation 3, and eight optimizer updates—exactly one pass over support24, without support cross-validation. Metrics are macro-F1, balanced accuracy, and NLL.

a) Cross-task evaluation.: We additionally evaluate InternVL3-8B and Qwen3-VL-8B on Camelyon17 hospital 2 (binary histopathology, support16/query300) and ManipBench Q1 (four-way vision–language action classification, query400), with three matched support seeds and identical query IDs for Frozen, rank-16 LoRA, Random-KV, and KV-TTT. Guardian/FailCoT uses InternVL3-8B on three fixed-semantics success/failure splits, support16, and three matched seeds. Corrected InternVL3 task results use the required `Answer: <label>` format; older raw-label ManipBench runs are excluded. Main performance results never use query labels. Mechanism diagnostics do, and are reported separately as oracles.

The evidence blocks serve different statistical roles. BRIGHT's primary four-method comparison uses one locked support24 per event; its independent three-support-seed suite estimates KV-versus-Frozen robustness but does not provide a matched multiseed LoRA comparison. Camelyon, Guardian, and ManipBench use matched support seeds. We keep these estimands separate rather than pooling incompatible replications.

B. Fixed-configuration main comparison

Table I uses one common rank-16, alpha-3, four-update KV configuration across events. Full improves the raw VLM on every event and has the best mean F1, while LoRA has the best mean NLL. Full exceeds fixed-8 LoRA on Libya and Turkey; LoRA remains better on Hawaii and Noto. Diagonal uses only 64 learned scalars but is less accurate. The table exposes the intended accuracy–adaptation-size trade-off rather than hiding metrics on which LoRA is better.

These architecture-transfer results reproduce the BRIGHT ranking without selecting a different actuator. Additional Phi, Gemma, and LLaVA controls are mixed—Ours is strongest on Gemma, LoRA on Phi, and Frozen on LLaVA—so the claim is cross-architecture evidence, not universal dominance.

C. Rank-16 alpha sweep

We sweep alpha in $\{0.5, 1, 2, 3, 5, 10\}$ for both controllers on all events (48/48 runs). Table IV reports per-event maxima to expose method capacity. These are *query-oracle diagnostics*, not valid primary results: alpha must be chosen using support-only evidence before query evaluation.

An independent-repeat warning is important: two nominally identical Hawaii rank-16 Full alpha-3 runs produce macro-F1 0.3295 and 0.2957 despite matching model fingerprint and recorded configuration. We therefore do not promote the oracle alpha-2 peak to the primary claim until repeated.

D. Mechanism controls and statistical robustness

At rank 5 and alpha 3, uncentered covariance reaches mean macro-F1 0.2964, versus 0.2506 for a rank-1 mean gradient, 0.2721 for centered covariance, and 0.2414 for a three-seed random basis. Ten-thousand-resample paired tests give covariance-minus-mean $+0.0458$ ($p = 0.0142$), covariance-minus-centered $+0.0243$ ($p = 0.0032$), and covariance-minus-random $+0.0551$ ($p = 0.0022$). One random seed approaches covariance, so the evidence supports higher and more reliable average performance, not failure of every random subspace.

Across three balanced support seeds and four events, compact Full versus raw VLM improves macro-F1 by $+0.0457$ (95% CI $[+0.0046, +0.0847]$, $p = 0.0167$); Diagonal improves by $+0.0399$ (95% CI $[+0.0063, +0.0734]$, $p = 0.0061$). Full versus Diagonal is not significant ($p = 0.7347$).

Table V places both performance and mechanism tests in the main paper. Confidence intervals and permutation tests operate on paired event and query outcomes; the random-basis comparison additionally samples the random basis seed within each bootstrap replicate. All reported improvements over raw and all covariance-mechanism contrasts reject a zero mean difference at the 5% level, whereas Full and Diagonal are statistically indistinguishable.

E. When does internal-state adaptation suffice?

Table VI combines primary performance with oracle mechanism evidence. It is not a correlation fit: the rows have deliberately different roles, and BRIGHT geometry uses archived support12 folds while its primary performance uses support24. Instead, the table implements a sequence of falsification tests for geometry transfer and actuator sufficiency.

a) Positive evidence-shift regime.: BRIGHT keeps the damage labels and decision semantics fixed while geography, sensor conditions, and damage appearance change. On the four-event locked Qwen2.5 protocol, Full KV obtains .3208 mean macro-F1 versus .2986 for LoRA and .2234 frozen, using 1,024 learned coefficients rather than 10.1M weight parameters. Qwen3-VL and InternVL3 show the same qualitative ordering in the four-method architecture-transfer table. Hawaii has strongly positive directional agreement in every class; Libya's aggregate agreement is weaker, but its class kappas all exceed .94.

b) Geometry transfers but realization can fail.: Camelyon has stable high ρ and κ across three seeds. Replacing the support basis by a query-derived basis while retaining the same KV actuator reaches .663 F1, comparable to or above the LoRA mean. The local actuator can therefore express the correction, while finite-support subspace estimation and coefficient-margin realization remain error sources. This is also visible in calibration: corrected InternVL KV-TTT has NLL .784 versus 1.990 for LoRA despite lower macro-F1, consistent with conservative probability correction rather than aggressive boundary movement.

c) Mixture and site mismatch.: RoboFail's class-conditional kappas are stable across seeds (success .962, failure .970), whereas aggregate kappa is seed-sensitive. Keeping the support covariance and basis fixed but reweighting support class means with the oracle query prior changes aggregate kappa from $.209 \pm .361$ to $.934 \pm .025$. The recovery occurs for Q, K, V, and O, directly implicating class-mixture cancellation. Q-only is the strongest seed-0 site diagnostic, but is not promoted to the method because it is not consistently beneficial elsewhere.

TABLE I

FIXED-CONFIGURATION COMPARISON. EVENT COLUMNS AND THEIR MEAN REPORT MACRO-F1; NLL IS AVERAGED OVER EVENTS (LOWER IS BETTER). KV USES RANK 16, LAYERS 14+27, ALPHA 3, AND FOUR UPDATES. "LEARNED" COUNTS EVENT-SPECIFIC OPTIMIZED SCALARS, EXCLUDING FROZEN COVARIANCE BASES.

Method	Hawaii	Libya	Noto	Turkey	Mean F1 ↑	Mean NLL ↓	Learned ↓
Raw VLM	0.1992	0.2585	0.2087	0.2273	0.2234	1.3233	0
LoRA fixed8	**0.3526**	0.2776	**0.3495**	0.2147	0.2986	**1.1276**	10,092,544
Diagonal KV	0.2918	0.2560	0.2820	0.2879	0.2794	1.1715	**64**
Full KV	0.2957	**0.3739**	0.3112	**0.3025**	**0.3208**	1.1367	1,024

TABLE II

FOUR-EVENT BRIGHT MEAN MACRO-F1 UNDER THE LOCKED SUPPORT24/QUERY300 PROTOCOL FOR TWO ADDITIONAL 8B BACKBONES.

Backbone	Frozen	LoRA	Random-KV	Ours
Qwen3-VL-8B	.1667	.2034	.1687	**.3492**
InternVL3-8B	.1848	.1667	.2214	**.2860**

d) Capability boundary.: ManipBench asks the model to compose instructions and images into changing A/B/C/D action semantics. Its directional agreement is positive, yet a query-derived KV basis reaches only .248 F1 and QKVO only .263, far below the .430 LoRA mean. Longer coefficient optimization improves ordinary KV but does not change this ordering. The bottleneck is therefore not merely finding the direction: local activation control lacks the functional authority of weight adaptation for task/action remapping.

F. Capacity, support sensitivity, and efficiency

Adapting every decoder layer with rank-16 Full matrices (14,336 scalars) is not beneficial. Naively retaining alpha 3 and learning rate 0.05 causes support-loss spikes and mean performance below the sparse controller. Reducing alpha to 0.3 and learning rate to 0.005 stabilizes Hawaii support loss, but query macro-F1 is only 0.2923 and damaged recall is zero. Sparse middle+final intervention is therefore empirically preferable to dense all-layer control.

Support selection also matters. Hawaii's original balanced support24 spans only three tiles. A representative-medoid alternative remains balanced and query-tile-disjoint but obtains 0.3084, below the original-support rank-5 alpha-2 value 0.3278. We report this as sensitivity, not as a replacement split.

On one RTX 3090 and the same 300 Hawaii queries, raw VLM, compact Full, and compact Diagonal require 1.3405, 1.3382, and 1.3384 seconds/sample; LoRA requires 1.3725 seconds/sample. Peak memory is 17,470 MiB for raw and KV and 17,508 MiB for LoRA. Compact Full and Diagonal learn 100 and 20 scalars (serialized states approximately 44 KiB), versus 10,092,544 parameters and 39.5 MiB for LoRA. KV extraction and coefficient fitting average 17 and 53 seconds.

V. LIMITATIONS

BRIGHT's clean four-method table uses one locked support set per event. Its three-seed robustness suite repeats KV but not LoRA, so we do not claim a matched multiseed KV–LoRA significance test. The archived BRIGHT geometry folds use support12 rather than the primary support24 and provide qualitative, not correlational, evidence. Query-basis, query-prior, and projection-site experiments use query labels and are mechanism oracles only. Small supports remain sensitive to spatial and class-mixture composition. The evaluation spans several VLM families, but not all backbones are run on every task. Finally, local KV control is intentionally not universal: it remains weaker than LoRA when the target requires substantial action/semantic remapping. These boundaries are part of the claim rather than targets for dataset- specific actuator selection.

VI. CONCLUSION

Gradient-covariance KV-TTT shows that a frozen VLM can sometimes be adapted with a tiny visual activation-state controller instead of a weight adapter. The key qualification is mechanistic: maximum support gradient-energy retention is only the first step. The correction must also transfer with the right sign, and the selected activation must have enough causal authority to realize it. BRIGHT illustrates the positive fixed-semantics evidence-shift regime; Camelyon separates actuator capacity from finite-support realization; RoboFail exposes class-mixture and site mismatch; and ManipBench establishes a task-remapping boundary. The resulting answer to "when can internal-state adaptation replace weight adaptation?" is neither always nor never: when the target correction is directionally transferable and locally expressible.

APPENDIX A
COMPLETE FIXED-CONFIGURATION METRICS

Table VII reports balanced accuracy (BA), NLL, and ordinal MAE in addition to the macro-F1 values in Table I. A lower NLL and MAE is better. The results show that improvements in macro-F1 need not imply an ordinal-error improvement: for example, fixed-8 LoRA raises Hawaii F1 while increasing MAE. This is why we retain both classification and probabilistic metrics rather than selecting a method from F1 alone.

APPENDIX B
COMPLETE RANK-16 ALPHA SWEEP

Tables VIII and IX contain all 48 runs behind the oracle summary. The non-monotonic response is important: increasing the residual bound does not consistently improve

TABLE III
CORRECTED INTERNVL3 CROSS-TASK MACRO-F1 (MEAN $\pm$ SAMPLE SD OVER THREE MATCHED SUPPORT SEEDS). GUARDIAN REPORTS THE FIXED EIGHT-STEP OURS FOLLOW-UP; RANDOM-KV WAS NOT RUN IN THAT SUITE.

Task	Frozen	LoRA	Random-KV	Ours
Camelyon17 hospital 2	$.3333 \pm .0000$	$.6325 \pm .2725$	$.4935 \pm .0071$	$.5107 \pm .0837$
Guardian UR5-Fail	$.4513 \pm .0067$	$.4432 \pm .0510$	–	$\mathbf{.4905 \pm .0308}$
Guardian RoboFail	$.4476 \pm .0263$	$\mathbf{.5010 \pm .0645}$	–	$.3952 \pm .0310$
Guardian RoboVQA	$.4043 \pm .0068$	$.5159 \pm .0087$	–	$\mathbf{.5204 \pm .0108}$
ManipBench droid-pick	$.1541 \pm .0000$	$\mathbf{.4304 \pm .2223}$	$.1556 \pm .0047$	$.2839 \pm .0729$

TABLE IV
PER-EVENT QUERY-ORACLE RANK-16 ALPHA MAXIMA.

Event	LoRA	Full	α	Diag. / α
Hawaii	0.3526	**0.3869**	2	0.3341 / 10
Libya	0.2776	**0.3852**	2	0.3390 / 5
Noto	**0.3495**	0.3270	1	0.3404 / 5
Turkey	0.2147	**0.3025**	3	0.2879 / 3
Mean	0.2986	**0.3504**	–	0.3254 / –

performance, and the best alpha differs by event and controller. Consequently, the maxima are used to diagnose capacity, not as the fixed primary estimator.

APPENDIX C
SUPERVISED ABLATIONS

Table X averages four events for the compact Full controller at alpha 3 while changing one factor at a time. The layer result is especially clear: middle+last substantially outperforms either layer alone. Rank 16 is the best mean in the original rank sweep, while eight updates slightly outperform four. Support48 does not improve on support24, consistent with support quality and spatial coverage mattering more than count alone.

APPENDIX D
SUPPORT-SEED ROBUSTNESS

Table XI summarizes the three balanced support24 seeds used by the compact alpha-3 controller. Variation is substantial, particularly on Noto Full and Libya Diagonal, motivating paired resampling over both support and queries. The statistical comparisons reported in the main paper use 10,000 stratified bootstrap and paired permutation replicates.

APPENDIX E
MECHANISM AND NEGATIVE CONTROLS

The mean-gradient control keeps only the normalized aggregate gradient and is therefore intrinsically rank one. Centered covariance subtracts the outer product of the token-level mean, while random controls use orthonormal Gaussian bases with three seeds. Their mean macro-F1 values are .2506, .2721, and .2414, respectively, versus .2964 for the uncentered second moment. Centering yields a similar NLL reduction (.1440 versus .1435) but lower F1, suggesting that both the gradient mean and directional variation contribute.

The dense all-layer negative control changes 56 projections. With rank 16 it has 14,336 coefficient scalars, but the original alpha-3/lr-.05 settings cause loss spikes and yield event F1 values .2400, .1670, .1704, and .1935. On Hawaii, alpha .3/lr .005 restores monotonically decreasing support loss yet yields only .2923 query F1 and zero damaged recall. This rules out the simple explanation that more controlled layers or more coefficient parameters necessarily help.

APPENDIX F
EFFICIENCY AND ARTIFACT ACCOUNTING

The timing excludes adaptation and measures identical query inference. KV basis extraction averages about 17 seconds and coefficient fitting about 53 seconds. The compact state files are approximately 44 KiB because they contain the bases as well as coefficients; the LoRA adapter is approximately 39.5 MiB. Full rank-16 has 1,024 learned coefficient scalars and Diagonal rank-16 has 64, but their exact end-to-end timing was not substituted for the measured compact benchmark in Table XII.

APPENDIX G
REPRODUCIBILITY AND SELECTION CAVEATS

All comparisons within a row use identical saved support and query manifests, model fingerprints, crop size, and query IDs. Completed predictions are configuration-bound and resumable. Nevertheless, two limitations affect interpretation. First, the current support sets are class-balanced but spatially concentrated (three Hawaii support tiles and four Noto support tiles). Second, two nominally matching Hawaii rank-16 Full alpha-3 runs differ by .0338 F1, despite matching recorded configuration. Independent repeats are therefore required for sensitive oracle peaks. Finally, no per-event alpha selected from the query sweep is presented as a test-independent primary result.

REFERENCES

[1] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021.

[2] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *NeurIPS*, 2017.

[3] M. Geva, R. Schuster, J. Berant, and O. Levy, "Transformer feed-forward layers are key-value memories," in *EMNLP*, 2021.

[4] Y. Sun, X. Wang, Z. Liu, J. Miller, A. A. Efros, and M. Hardt, "Test-time training with self-supervision for generalization under distribution shifts," in *ICML*, 2020.

TABLE V

PAIRED STATISTICAL VALIDATION (10,000 RESAMPLES). POSITIVE Δ FAVORS THE FIRST METHOD. MECHANISM ROWS USE THE FIXED SEED-0 RANK-5 CONTROLLER; RAW COMPARISONS SAMPLE THREE SUPPORT SEEDS.

Comparison	Mean Δ Macro-F1	95% bootstrap CI	Permutation p
Full KV $-$ raw VLM	$+0.0457$	$[+0.0046, +0.0847]$	0.0167
Diagonal KV $-$ raw VLM	$+0.0399$	$[+0.0063, +0.0734]$	0.0061
Full KV $-$ Diagonal KV	$+0.0058$	$[-0.0433, +0.0506]$	0.7347
Covariance $-$ mean gradient	$+0.0458$	$[+0.0188, +0.0735]$	0.0142
Covariance $-$ centered covariance	$+0.0243$	$[+0.0100, +0.0388]$	0.0032
Covariance $-$ random basis	$+0.0551$	$[+0.0058, +0.1036]$	0.0022

TABLE VI

GEOMETRY–ACTUATOR REGIMES. ΔF1 IS OURS MINUS LORA UNDER THE CORRESPONDING CLEAN PRIMARY PROTOCOL. QUERY-BASIS AND SITE RESULTS ARE QUERY-LABEL ORACLES, NOT DEPLOYABLE SCORES.

Task	ρ	κ	ΔF1	Same-actuator oracle	Diagnosis
BRIGHT Hawaii	.433	.975	+.171	–	fixed-semantics evidence shift
BRIGHT Libya	.369	.677	+.058	–	transferable class geometry
Camelyon17	$.785 \pm .005$	$.977 \pm .008$	-.122	query-basis KV .663 vs LoRA .633	finite-support/margin gap
RoboFail	$.620 \pm .008$	$.209 \pm .361$	-.106	Q-only .481 vs LoRA .501	mixture and site mismatch
ManipBench droid-pick	$.620 \pm .003$	$.741 \pm .142$	-.147	query-basis KV .248 vs LoRA .430	actuator/capability boundary

TABLE VII

COMPLETE METRICS FOR THE FIXED RANK-16, ALPHA-3 COMPARISON.

Event	Method	Macro-F1	BA	NLL	Ordinal MAE
Hawaii	Raw / LoRA / Full / Diag.	0.1992 / 0.3526 / 0.2957 / 0.2918	0.3433 / 0.3767 / 0.3533 / 0.3300	1.2491 / 1.0974 / 1.1314 / 1.1235	0.6767 / 0.8600 / 0.9467 / 0.7800
Libya	Raw / LoRA / Full / Diag.	0.2585 / 0.2776 / 0.3739 / 0.2560	0.3633 / 0.3467 / 0.3800 / 0.3567	1.3331 / 1.1175 / 1.1011 / 1.1525	0.6633 / 0.9200 / 0.8167 / 0.6667
Noto	Raw / LoRA / Full / Diag.	0.2087 / 0.3495 / 0.3112 / 0.2820	0.3333 / 0.3800 / 0.3400 / 0.3500	1.2779 / 1.0977 / 1.1467 / 1.1405	0.6867 / 0.8933 / 0.8767 / 0.6967
Turkey	Raw / LoRA / Full / Diag.	0.2273 / 0.2147 / 0.3025 / 0.2879	0.3400 / 0.3233 / 0.3067 / 0.3400	1.4333 / 1.1980 / 1.1676 / 1.2694	0.7300 / 1.0233 / 0.9133 / 0.8733

TABLE VIII

FULL RANK-16 MACRO-F1 BY ALPHA.

Event	.5	1	2	3	5	10
Hawaii	.2622	.3338	**.3869**	.2957	.3239	.3476
Libya	.3249	.2760	**.3852**	.3739	.3394	.2943
Noto	.2354	**.3270**	.3095	.3112	.3021	.2770
Turkey	.2627	.1876	.2527	**.3025**	.2472	.2578

TABLE IX

DIAGONAL RANK-16 MACRO-F1 BY ALPHA.

Event	.5	1	2	3	5	10
Hawaii	.2880	.2972	.3021	.2918	.3211	**.3341**
Libya	.3110	.3033	.3041	.2560	**.3390**	.3278
Noto	.2629	.2677	.2278	.2820	**.3404**	.3310
Turkey	.2637	.2698	.2604	**.2879**	.2843	.2601

TABLE X

FOUR-EVENT MEAN MACRO-F1 FOR ONE-FACTOR SUPERVISED ABLATIONS.

Factor	Setting	Mean F1
Support	12 / 24 / 48	.2538 / **.2964** / .2754
Layers	14 / 27 / 14+27	.1964 / .2213 / **.2964**
Rank	5 / 8 / 16 / 32	.2964 / .2972 / **.3136** / .2930
Updates	1 / 2 / 4 / 8	.2486 / .2219 / .2964 / **.3048**
Learning rate	.01 / .05 / .1 / .2	.2659 / **.2964** / .2513 / .2466

TABLE XI

MACRO-F1 MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THREE SUPPORT SEEDS.

Event	Full	Diagonal
Hawaii	$.2795 \pm .0317$	$.2883 \pm .0344$
Libya	$.3044 \pm .0223$	$.3111 \pm .0530$
Noto	$.2611 \pm .0566$	$.1982 \pm .0116$
Turkey	$.2317 \pm .0498$	$.2557 \pm .0293$

[5] D. Wang, E. Shelhamer, Z. Liu, K. Olshausen, and T. Darrell, "Fully test-time adaptation by entropy minimization," in *ICLR*, 2021.

[6] Q. Wang, O. Fink, L. Van Gool, and D. Dai, "Continual test-time domain adaptation," in *CVPR*, 2022.

[7] M. Shu, W. Nie, D.-A. Huang, Z. Yu, T. Goldstein, A. Anandkumar, and C. Xiao, "Test-time prompt tuning for zero-shot generalization in vision–language models," in *NeurIPS*, 2022.

[8] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, "Parameter-efficient transfer learning for NLP," in *ICML*, 2019.

[9] B. Lester, R. Al-Rfou, and N. Constant, "The power of scale for parameter-efficient prompt tuning," in *EMNLP*, 2021.

[10] K. Meng, A. Sharma, A. Andonian, Y. Belinkov, and D. Bau, "Locating and editing factual associations in GPT," in *NeurIPS*, 2022.

[11] K. Meng, D. Bau, A. Adrian, and Y. Belinkov, "Mass-editing memory in a transformer," in *ICLR*, 2023.

[12] R. Gupta, R. Hosfelt, S. Sajeev, and et al., "xbd: A large-scale dataset for building damage assessment," in *NeurIPS Datasets and Benchmarks*, 2021.

TABLE XII
EXACT HAWAII QUERY300 EFFICIENCY ON ONE RTX 3090.

Method	sec/sample	Peak MiB	Trainable
Raw VLM	1.3405	17,470	0
LoRA	1.3725	17,508	10,092,544
Full rank5	1.3382	17,470	100
Diagonal rank5	1.3384	17,470	20

[13] I. Bumbacher and et al., "Bright: Benchmark for rapid and insightful and systemic disaster response from satellite imagery," in *NeurIPS Datasets and Benchmarks*, 2024.

[14] Y. Ravi and et al., "Satmae: Pre-training transformers for temporal and multi-spectral satellite imagery," in *NeurIPS*, 2023.

[15] M. Bhatia and et al., "Foundation models for rapid disaster response reasoning," in *ACL Findings*, 2024.

[16] S. Bai, K. Bai, and et al., "Qwen2.5-vl technical report," *arXiv preprint arXiv:2502.13923*, 2025.

[17] K. Fan, "Maximum properties and inequalities for the eigenvalues of completely continuous operators," *Proceedings of the National Academy of Sciences*, vol. 37, no. 11, pp. 760–766, 1951.